

College Database – SQL Practice

3 Tables • Sample Values • WHERE • GROUP BY • HAVING • LIMIT • ORDER BY • JOIN • Subqueries

1. Create Database

```
CREATE DATABASE college_db;
USE college_db;
```

2. Create Tables – Students

```
CREATE TABLE students (
    student_id INT PRIMARY KEY,
    student_name VARCHAR(50),
    age INT,
    city VARCHAR(50),
    course_id INT
);
```

3. Create Tables – Courses

```
CREATE TABLE courses (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(50),
    fees INT
);
```

4. Create Tables – Marks

```
CREATE TABLE marks (
    mark_id INT PRIMARY KEY,
    student_id INT,
    subject VARCHAR(50),
    marks INT
);
```

5. Insert Values – Students

```
INSERT INTO students VALUES
(1, 'Aarav', 20, 'Pune', 101),
(2, 'Sneha', 21, 'Mumbai', 102),
(3, 'Rohan', 19, 'Nashik', 101),
(4, 'Priya', 22, 'Pune', 103),
(5, 'Kunal', 20, 'Mumbai', 102),
(6, 'Neha', 21, 'Nashik', 103),
(7, 'Aditya', 19, 'Pune', 101),
(8, 'Pooja', 22, 'Mumbai', 103);
```

6. Insert Values – Courses

```
INSERT INTO courses VALUES
(101, 'Computer Science', 60000),
(102, 'Data Science', 70000),
(103, 'Information Technology', 55000);
```

7. Insert Values – Marks

```
INSERT INTO marks VALUES
(1, 1, 'SQL', 85),
(2, 1, 'Python', 90),
```

```
(3, 2, 'SQL', 78),  
(4, 2, 'Python', 88),  
(5, 3, 'SQL', 92),  
(6, 3, 'Python', 81),  
(7, 4, 'SQL', 75),  
(8, 4, 'Python', 79),  
(9, 5, 'SQL', 89),  
(10, 5, 'Python', 94),  
(11, 6, 'SQL', 70),  
(12, 6, 'Python', 76),  
(13, 7, 'SQL', 95),  
(14, 7, 'Python', 91),  
(15, 8, 'SQL', 82),  
(16, 8, 'Python', 87);
```

SQL Practice Questions

WHERE

1. Display all students who are from Pune.
2. Display students whose age is greater than 20.
3. Display students whose age is 20 or less.
4. Display students who are from Mumbai and age is greater than 20.
5. Display marks where the marks are greater than 85.
6. Display students whose name starts with 'P'.

ORDER BY

7. Display all students sorted by age in ascending order.
8. Display all students sorted by age in descending order.
9. Display marks sorted from highest to lowest.
10. Display courses sorted by fees from lowest to highest.

LIMIT

11. Display the first 3 students.
12. Display the top 5 highest marks.
13. Display the 2 students with the highest age.
14. Display the cheapest 2 courses.

GROUP BY

15. Find the number of students in each city.
16. Find the number of students in each course.
17. Find the average marks for each subject.
18. Find the maximum marks for each subject.
19. Find the minimum marks for each subject.

HAVING

20. Find cities having more than 2 students.
21. Find courses having more than 2 students.
22. Find subjects where the average marks are greater than 80.
23. Find cities where the number of students is at least 3.

JOIN

24. Display student name and their course name.
25. Display student name, city, and course name.

26. Display student name and their marks.
27. Display student name, subject, and marks.
28. Display student name, course name, and course fees.
29. Display students who are enrolled in Data Science.
30. Display students who scored more than 85, along with their names.

SUBQUERIES

31. Find students whose marks are greater than the average marks.
32. Find the student who has the highest marks.
33. Find the student who has the lowest marks.
34. Find courses whose fees are greater than the average course fees.
35. Find students who are enrolled in the most expensive course.
36. Find students who belong to the same course as Aarav.
37. Find students whose marks are greater than Rohan's SQL marks.
38. Find the course with the highest fees.

MIXED PRACTICE

39. Display the top 3 students with the highest marks.
40. Find the average marks of each student and display only students whose average is greater than 85.
41. Display each city and the number of students, sorted by the number of students from highest to lowest.
42. Display the top 2 courses based on fees.
43. Display course name and number of students in each course, but show only courses having more than 2 students.
44. Display the names of students who scored above the overall average marks, along with their course names.
45. Find the student with the highest Python marks and display their name, marks, and course name.